CS3211 Concurrent Programming

AY 25/26 Sem 2 — github/omgeta

1. Introduction

Concurrency refers to a system performing multiple independent tasks in overlapping time periods.

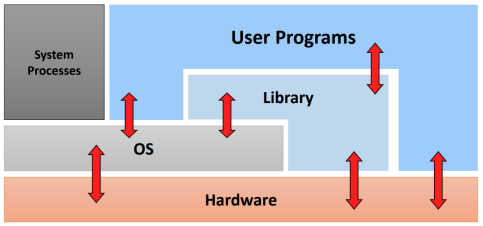
Concurrency vs. Parallelism:

i. Concurrency: tasks progress in overlapping time
ii. Parallelism: tasks also execute simultaneously
   - Enabled by multiple cores, each can have multiple hardware threads (SMT)
   - Hardware threads dictate parallel execution of kernel threads

Program Parallelization Steps:

i. Decomposition: split problem into parallel tasks
   - Granularity: size of task
ii. Scheduling: assign tasks to threads
iii. Orchestration: synchronisation and communication of tasks to satisfy dependencies
iv. Mapping: bind threads to physical cores

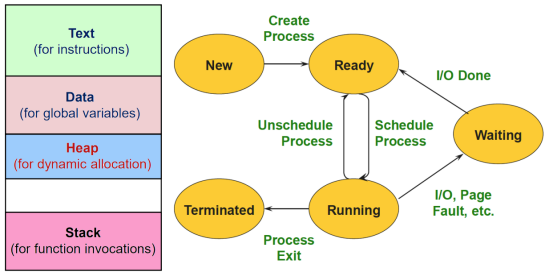
Software Stack:



Processes

Process is an abstraction for a running program:

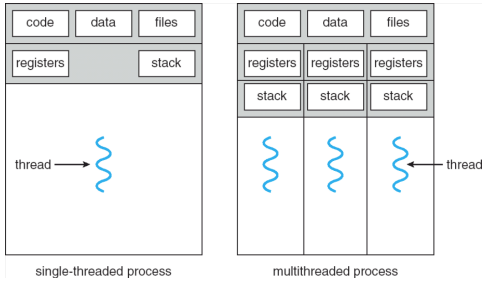
– High system call overhead
– OS allocates and initializes data structures
– Communication costly through the OS



Threads

Threads are independent execution flows within a process:

i. Shared: Resources, Address Space
ii. Private: Thread ID, Registers, "Stack" (diff. SP)
+ Efficient: cheaper creation and context switching
+ Resource Sharing: process resources can be shared



Implementations:

i. User-Threads are managed by a userspace library:
   + Fast: context switches have low overhead
   – Cannot parallelize to different cores
   – Threads blocking will block whole process
ii. Kernel-Threads are implemented in the OS:
   + Parallel: threads can run across many cores
   – Slower: thread operations are system calls

Synchronisation

Race Condition:

i. Program outcome depends on timing/ordering of concurrent operations

Data Races lead to undefined behaviour:

i. Two concurrent access to same memory location
ii. At least one access is a write
iii. No synchronization

Mutual Exclusion synchronises access to shared resources:

i. Thread locks mutex to access shared resources
ii. Same thread unlocks mutex once shared resources no longer used

Symptoms of Incorrect Synchronization:

i. Deadlock: there is an execution where no progress is made when it should be possible
   - Indistinguishable from livelock
ii. Starvation: a thread cannot progress because other threads hold a required resource, which it does not get a chance to acquire

Mechanisms:

i. Lock: primitive `acquire()`, `release()`
ii. Semaphore: atomic ≥ 0; `acquire()`, `release()`
   - Binary: represents single access
   - Counting: number of threads in CS $\leq N$
   - Can be manipulated by different threads
   - No relation to data being controlled
iii. Monitors: high-level implicit concurrency
iv. Messages: explicit model of communication and synchronisation based on atomic data transfer

2. C++

Threads

Operations:

i. Construct:
   - Function: `std::thread t(f, args...)`
   - Callable object: `std::thread t{F{}, args...}`
   - Lambda: `std::thread t([]{ ... })`

ii. Wait Once: `t.join()`
   - Check: `t.joinable()`
   - Must still join on exception paths

iii. Detach: `t.detach()`
   – Detached thread no longer joinable
   – May outlive local variable references

iv. Pass By Reference: `std::ref(x)`

v. Ownership: movable, not copyable

Mutexes

Operations:

i. Construct: `std::mutex m`

ii. Explicit: `m.lock()`, `m.unlock()`
   – Must unlock on every exit path, incl. exceptions

iii. RAII: `std::lock_guard<std::mutex> g(m)`
   + Locks on construction and unlocks on scope exit

iv. Flexible: `std::unique_lock<std::mutex> lk(m)`
   - Adopt Existing Lock: `lk(m, std::adopt_lock)`
   - Defer Lock: `lk(m, std::defer_lock)`
   + Unlocks on scope exit if mutex owned
   + Allows multiple lock/unlock within a scope
   + Needed for condition variables
   + Moveable: can transfer lock ownership

v. Multiple Mutexes (Deadlock-Free):
   - Lock in total order, avoid nested locks
   - Explicit: `std::lock(m1, m2, ...)`
   - RAII: `std::scoped_lock lk(m1, m2, ...)`

Condition Variables

Condition variables sync actions without busy-waiting.

Operations:

i. Construct: `std::condition_variable cv`
   - Works only on `std::unique_lock<std::mutex>`
   - Generic (any mutex-like, less efficient): `std::condition_variable_any cva`

ii. Wait: `cv.wait(lk)` / `cv.wait(lk, pred)`
   - Checks predicate with mutex locked
   - If condition is met, returns
   - If condition is unmet, unlocks mutex and blocks
   – Spurious Wake: may wake without notification to check condition; so should be side effect-free

iii. Wake One Thread: `cv.notify_one()`

iv. Wake All Threads: `cv.notify_all()`

Monitors

Monitors encapsulate shared data and operations enforced by a mutex and condition variables.

i. One thread executes monitor operations at a time

ii. Threads wait for conditions and state changes

Example:

```
class JobQueue {
  std::mutex m;
  std::condition_variable cv;
  std::queue<Job> q;
public:
  void enqueue(Job j) {
    {
      std::unique_lock<std::mutex> lk(m);
      q.push(j);
    }
    cv.notify_one();
  }

  Job dequeue() {
    std::unique_lock<std::mutex> lk(m);
    cv.wait(lk, [&]{ return !q.empty(); });
    Job j = q.front(); q.pop();
    return x;
  }
};
```

Memory Model

Memory model is the contract about the visible side effects of memory operations.

i. As-if: compiler/hardware may reorder operations if single-threaded observable behavior is unchanged.

Ordering Relations:

i. Sequenced-Before: program order

ii. Synchronizes-With: suitably-tagged store → load on same atomic, if the load reads that value

iii. Happens-Before: transitive closure of Sequenced-Before ∪ Synchronizes-With

iv. Modification Order: per-atomic total order of writes consistent across threads

v. Coherence Order: per-atomic partial order of writes (totally ordered) and reads (unordered w.r.t others) consistent across threads

vi. Visible Side-Effect: write visible to a read

Atomics (`std::atomic<T>`) have indivisible operations:

i. Load: `a.load(std::memory_order)`

ii. Store: `a.store(val, std::memory_order)`

iii. RMW: `a.fetch_add(val, std::memory_order)`, `a.exchange(val, std::memory_order)`

iv. May be lock-free/implemented with mutex: `a.is_lock_free()`, `T::is_always_lock_free`

Memory Models:

i. Sequentially Consistent (`seq_cst`):
   - All threads agree on same order of operations

ii. Relaxed (`relaxed`):
   - No Synchronizes-With cross-thread guarantees

iii. Acquire-Release (`acquire`, `release`, `acq_rel`)
   - Release store → Acquire load on same atomic can establish Synchronizes-With
   - Writes before release become visible after the acquire (in the acquiring thread)
   - Release Sequence: chain of RMWs on the same atomic after release; an acquire reading from the chain synchronizes-with the original release

Lock-based Data Structures

Lock-based structures protect shared state with mutexes.

i. Lock the right mutex before accessing protected data
ii. Hold locks for the minimum time
iii. Interface should not expose race conditions
iv. Deadlocks are possible

Stack (one global mutex)

```
void push(T x) {
  std::lock_guard<std::mutex> lk(m);
  data.push(std::move(x));
}

std::shared_ptr<T> pop() {
  std::lock_guard<std::mutex> lk(m);
  if (data.empty()) throw empty_stack();
  auto res =
    std::make_shared<T>(std::move(data.top()));
  data.pop();
  return res;
}
```

+ Safe for concurrent calls, exception-safe
− Work is serialized
− No built-in waiting for items: must poll `empty()` or catch `empty_stack` exceptions

Queue (one mutex + notification)

```
void push(T x) {
  std::lock_guard<std::mutex> lk(mut);
  data_queue.push(std::move(x));
  data_cond.notify_one();
}

void wait_and_pop(T& value) {
  std::unique_lock<std::mutex> lk(mut);
  data_cond.wait(lk, [this]{ return
    !data_queue.empty(); });
  value = std::move(data_queue.front());
  data_queue.pop();
}
```

+ `wait_and_pop()` blocks until an item is available
+ Safe for concurrent member calls
− One mutex still serializes queue operations
− `notify_one()` + exception in `wait_and_pop()` can leave other waiters asleep

Queue (push shared-pointers in the queue)

```
void push(T x) {
  auto data = std::make_shared<T>(std::move(x));
  std::lock_guard<std::mutex> lk(mut);
  data_queue.push(data);
  data_cond.notify_one();
}

std::shared_ptr<T> wait_and_pop() {
  std::unique_lock<std::mutex> lk(mut);
  data_cond.wait(lk, [this]{ return
    !data_queue.empty(); });
  auto res = data_queue.front();
  data_queue.pop();
  return res;
}
```

+ Data can be allocated before locking
+ Returned value can share ownership with the queue element
− One mutex still serializes queue operations

Queue (fine-grained; dummy node)

```
node* get_back() {
  std::lock_guard<std::mutex> lk(back_mutex);
  return back;
}

std::unique_ptr<node> pop_front() {
  std::lock_guard<std::mutex> lk(front_mutex);
  if (front.get() == get_back()) return nullptr;
  auto old_front = std::move(front);
  front = std::move(old_front->next);
  return old_front;
}

void push(T x) {
  auto new_data =
    std::make_shared<T>(std::move(x));
  auto p = std::make_unique<node>();
  node* new_back = p.get();
  std::lock_guard<std::mutex> lk(back_mutex);
  back->data = new_data;
  back->next = std::move(p);
  back = new_back;
}
```

+ Separate `front_mutex` and `back_mutex`
+ `push()` only locks the back
+ More concurrency than one global mutex
− Locking protocol and empty-check become more complex

Queue (more fine-grained)

```
std::unique_ptr<node> try_pop_head() {
  std::unique_lock<std::mutex>
    front_lk(front_mutex);
  if (front.get() == get_back()) return nullptr;

  std::unique_lock<std::mutex>
    node_lk(front->mutex);
  auto old_head = std::move(front);
  front = std::move(old_head->next);
  return old_head;
}

std::shared_ptr<T> try_pop() {
  std::unique_ptr<node> old_head;
  { old_head = try_pop_head(); }   // lock
    released before delete
  return old_head ? old_head->data :
    std::shared_ptr<T>();
}

void push(T x) {
  auto new_data =
    std::make_shared<T>(std::move(x));
  auto p = std::make_unique<node>();
  node* new_back = p.get();

  std::lock_guard<std::mutex> back_lk(back_mutex);
  std::lock_guard<std::mutex>
    node_lk(back->mutex);
  back->data = new_data;
  back->next = std::move(p);
  back = new_back;
}
```

+ Per-node mutex
+ More fine-grained synchronization between `push()` and `pop()`
− More complex locking protocol
− Must release locks before destroying the old node

Lock-Free Data Structures

Lock-free structures avoid blocking library calls.

i. Obstruction-free: a thread completes if it eventually runs in isolation
ii. Lock-free: from any point, some thread completes
   - Livelock-free but Starvation possible
   - \+ Maximum concurrency: some thread progresses
   - \+ Robust: if one thread dies, others can proceed
   - – Atomics may be much slower than non-atomics
   - – Contention/write propagation hurt performance
   - – Cache ping-pong and false sharing
iii. Wait-free: every running thread completes regardless of execution speed of others
   - Livelock- and Starvation-free; stronger lock-free
   - \+ Per-thread progress guaranteed
   - – Harder to design and prove correct
   - – Often higher overhead than lock-free structures

Risks:

i. Livelock: from unbounded restarts in response to other thread actions
ii. Starvation: loops with CAS, or algorithms with unbounded number of retries due to clashes

Guidelines:

i. Start with `std::memory_order_seq_cst`
ii. Use CAS loops carefully
iii. Watch for ABA
iv. Use safe memory reclamation
v. Prefer lock-based unless lock-free progress needed

CAS Loop Pattern

```
int fetch_add(std::atomic<int>& value, int add) {
  int old_value = value.load();
  while (true) {
    int new_value = old_value + add;
    if (value.compare_exchange_weak(old_value,
    new_value))
      return old_value;
  }
}
```

Queue (initial idea)

```
void push(Job job) {
  Node* new_dummy = new Node();
  Node* work_node =
    m_queue_back.exchange(new_dummy);
  work_node->job = job;
  work_node->next.store(new_dummy);
}

std::optional<Job> try_pop() {
  Node* old_front = m_queue_front.load();
  while (true) {
    Node* new_front = old_front->next.load();
    if (new_front == QUEUE_END)
      return std::nullopt;
    if (m_queue_front.CAS(old_front, new_front))
      return old_front->job;
  }
}
```

- `exchange()` gives each producer its exclusive dummy node
- `try_pop()` advances the head with a CAS loop

Issue 1: Data races

```
Node* work_node =
    m_queue_back.exchange(new_dummy,
    stdmo::acq_rel);
work_node->next.store(new_dummy, stdmo::release);

Node* old_front =
    m_queue_front.load(stdmo::acquire);
Node* new_front =
    old_front->next.load(stdmo::acquire);
m_queue_front.compare_exchange_weak(old_front,
    new_front, stdmo::acq_rel);
```

- Producer vs. producer: `m_queue_back.exchange(..., stdmo::acq_rel)`
- Producer vs. its consumer: release store to `next`, acquire load from `next`
- Producer vs. next consumer: acquire / acq_rel accesses on `m_queue_front`

Issue 2: ABA

```
struct alignas(16) GenNodePtr {
  Node* node;
  uintptr_t gen;
};

std::atomic<GenNodePtr> m_queue_front;
```

- CAS may succeed even if a pointer changes away from A and back to A
- Fix: add a generation counter to the front pointer

Issue 3: Use-after-free

```
Node* get_recycled_node_or_allocate_new();
void add_node_to_recycling_stack(Node* node);
```

- Old front pointers may become stale
- Fix: do not delete nodes directly; reuse nodes via a lock-free recycling stack

Issue 4: Recycling race

```
Node* cur_stack_next =
  old_stack_top.node->next.load(stdmo::acquire);

node->next.store(old_stack_top.node,
    stdmo::release);
```

- Recycling also needs synchronization
- Use release / acquire on recycled-node `next`

Issue 5: Serializability

- The repaired queue can still be non-serializable: head / tail may be updated before fully linked
- Fix: Michael-Scott Queue prioritizes maintaining a valid linked list

3. Classical Synchronisation Problems

Producer-Consumer

Processes share a bounded buffer of fixed size K, where producers add items until buffer full and consumers remove items when not empty.

Blocking Solution (init `not_full = K, not_emp = 0`):

```
// Producer
x = produce();

wait(not_full);
wait(mutex);
buffer.add(x);
signal(mutex);
signal(not_emp);
```

```
// Consumer
wait(not_emp);
wait(mutex);
x = buffer.get();
signal(mutex);
signal(not_full);

consume(item);
```

Sleeping Barber

Barber sleeps until a customer wakes him. If barber is busy and chairs are free, the customer sits or will leave if there are no chairs.

```
// Customer
wait(mutex);
if (customers==n){
  signal(mutex);
  exit();
}
customers += 1;
signal(mutex);

signal(customer);
wait(barber);
getHairCut();
signal(custDone);
wait(barbDone);

wait(mutex);
customers -= 1;
signal(mutex);
```

```
// Barber
while (1) {
  wait(customer);
  signal(barber);
  cutHair();
  wait(custDone);
  signal(barbDone);
}
```

Reader-Writer

Processes share a critical region, where readers can read simultaneously but writers must have exclusive access.

```
// Writer
wait(empty);
write();
signal(empty);
```

```
// Reader
wait(mutex);
readers++;
if (readers == 1)
  wait(empty);
signal(mutex);

read();

wait(mutex);
readers--;
if (readers == 0)
  signal(empty);
signal(mutex);
```

- Writer Starvation: possible if readers keep entering and `empty` is never signalled. Solutions:
  - Queue: all pass through `wait(queue)`
  - Writer-Priority: readers entering must `wait(readTry)`; but first waiting writer blocks new readers with `wait(readTry)` and does `signal(readTry)` when done

Barrier

```
int i_am_last = 0;
wait(mutex);
count++;
if (count == N) {
  i_am_last = 1;
  signal(barrier);
}
signal(mutex);

wait(barrier);
if (!i_am_last)
  signal(barrier);
```

Dining Philosophers

N philosophers around a circular table, with a single chopstick between. 2 chopsticks are needed to eat.

```
// Philosopher
void p(int i) {
  while (1) {
    // think
    take(i);
    eat();
    put(i);
  }
}

void safe_to_eat
   (int i) {
  if (state[i] ==
   HUNGRY &&
   state[LEFT] !=
   EATING &&
   state[RIGHT] !=
   EATING) {
    state[i] =
   EATING;
    signal(s[i]);
  }
}
```

```
// Take Chopstick
void take(int i) {
  wait(mutex);
  state[i] =
   HUNGRY;
  safe_to_eat(i);
  signal(mutex);
  wait(s[i]);
}

// Put Chopstick
void put(int i) {
  wait(mutex);
  state[i] =
   THINKING;

   safe_to_eat(LEFT);

   safe_to_eat(RIGHT)
  signal(mutex);
}
```

Limited Eater Solution, when at most $N-1$ philosophers eat concurrently, it's guaranteed at least one can eat:

```
void philosopher(int i) {
  while (1) {
    // think
    wait(seats);
    wait(chopstick[LEFT]);
    wait(chopstick[RIGHT]);
    eat();
    signal(chopstick[RIGHT]);
    signal(chopstick[LEFT]);
    signal(seats);
  }
}
```

4. Testing

Bug Types:

i. Unwanted Blocking:
   - Deadlock/Livelock: no progress, operations are not "completed", effects do not become visible
   - IO: waiting for external input

ii. Race Conditions:
   - Data races ⇒ undefined behavior (unsynchronized concurrent access)
   - Broken invariants:
     - Double-free
     - Dangling pointers $\implies$ UAF
     - Memory corruption from read partial update
   - Lifetime issues: thread outlives data it accesses

Techniques:

i. Review the code:
   - Identify shared data and how it is protected
   - Track held mutexes and find lock ordering issues
   - Check cross-thread ordering enforcement
   - Validate lifetimes (e.g. `join()` on exceptions)

ii. Testing:
   - Run the smallest amount of code that can demonstrate the problem
   - Remove concurrency to confirm it is concurrency-related
   - Run on multicore and (when useful) single-core

Dynamic Instrumentation

Valgrind:

i. Shadows registers and memory to validate memory operations:
   each allocation freed once, each access to allocated space, each reach from initialized memory

ii. High overhead: `10-20x`

iii. Usage: `valgrind --tool=memcheck <prog ...>`

Helgrind:

i. Intercepts `pthread` operations to validate thread operations:
   miuse of POSIX threads API, potential deadlocks from lock-order cycles, data races from happens-before analysis

ii. Very high overhead: `~100x`

iii. Usage: `valgrind --tool=helgrind <prog ...>`

Compiler Instrumentation

Sanitizers:

i. Compile-time instrumentation; supported by `clang` and `gcc`

ii. Lower overhead: `~5-10x`

iii. Instruments memory accesses / function boundaries and tracks memory state

AddressSanitizer (ASan):

i. Detects addressability issues

ii. Typical cost: `~2x` slowdown, `1.5-3x` memory overhead

iii. Usage: `-fsanitize=address`

ThreadSanitizer (TSan):

i. Detects data races using happens-before tracking

ii. Also reports thread / mutex / lock-order issues

iii. Typical cost: `5-15x` slowdown, `5-10x` memory overhead

iv. Usage: `-fsanitize=thread -fPIE -pie`

Appendix

Classes

Class Skeleton:

```
class C {
  std::string x;
public:
  C(): x("default") {
    // default constructor
  }

  C(int i): x(std::string(i)) {
    // custom constructor
  }

  C(const C& other) : x(other.x) {
    // copy constructor
  }

  C& operator=(const C& other) {
    // copy assignment operator
    x = other.x;
    return *this;
  }

  C(C&& other): x(std::move(other.x)) {
    // move constructor
  }

  C& operator=(C&& other) {
    // move assignment operator
    x = std::move(other.x);
    return *this;
  }

  ~C() {
    // destructor
  }
};
```

Rule of 3: if a class needs a custom destructor, copy constructor, or copy assignment operator then it usually needs all 3

Smart Pointers

Smart pointers manage heap resources automatically.

Ownership:

i. `std::unique_ptr<T>`: exclusive ownership
ii. `std::shared_ptr<T>`: shared ownership via atomic reference counting
iii. `std::weak_ptr<T>`: non-owning observer of a `shared_ptr`

Basic Methods:

i. Construct:
   - `auto p = std::make_unique<T>(args...)`
   - `auto p = std::make_shared<T>(args...)`
ii. Access:
   - Dereference: `*p`, `p->member`
   - Raw pointer: `p.get()`
iii. Reset / replace:
   - `p.reset()`
   - `p.reset(new T(...))`
iv. Move ownership:
   - `std::unique_ptr` is movable, not copyable
   - `auto q = std::move(p)`
v. Shared count:
   - `p.use_count()`
   - Copying a `shared_ptr` increments the count
vi. Weak access:
   - `w.lock()` gives a `shared_ptr` if object still exists
   - `w.expired()` checks whether object has already been destroyed
vii. Unique-pointer release:
   - `p.release()` returns raw pointer and gives up ownership